

**Project Design Phase-II**  
**Solution Requirements (Functional & Non-functional)**

|               |                                                     |
|---------------|-----------------------------------------------------|
| Date          | 29 June 2026                                        |
| Name          | HARRI VENDHAN L                                     |
| Project Name  | <b>Metro Ticket Generating System in ServiceNow</b> |
| Maximum Marks | 4 Marks                                             |

**Functional Requirements**

Following are the functional requirements of the proposed **Metro Ticket Booking System**.

| FR No. | Functional Requirement (Epic) | Sub Requirement (Story / Sub-Task)                                                                                                                                                                                                    |
|--------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-1   | Metro Ticket Booking          | <ul style="list-style-type: none"><li>• Allow passengers to select the journey type, source station, destination station, and number of passengers.</li><li>• Calculate the fare automatically based on the selected route.</li></ul> |
| FR-2   | Smart Card Recharge           | <ul style="list-style-type: none"><li>• Allow users to enter their smart card number and recharge amount.</li><li>• Store recharge details securely in the Metro Database.</li></ul>                                                  |
| FR-3   | QR Code Ticket Generation     | <ul style="list-style-type: none"><li>• Automatically generate a unique QR Code after successful ticket booking.</li><li>• Store the generated QR Code URL in the Metro Database for future verification.</li></ul>                   |
| FR-4   | Booking & Payment Management  | <ul style="list-style-type: none"><li>• Support multiple payment methods (Card, UPI, Net Banking, etc.).</li><li>• Save booking details, passenger information, fare amount, and payment information in the Metro Database.</li></ul> |
| FR-5   | Record Management             | <ul style="list-style-type: none"><li>• Create booking records automatically using ServiceNow Flow Designer.</li><li>• Maintain booking history and transaction details for future reference.</li></ul>                               |
| FR-6   | User Notification             | <ul style="list-style-type: none"><li>• Display booking confirmation after successful submission.</li><li>• Provide passengers with the generated QR ticket for metro travel.</li></ul>                                               |

## Non-Functional Requirements

Following are the non-functional requirements of the proposed **Metro Ticket Booking System**.

| <b>NFR No.</b> | <b>Non-Functional Requirement</b> | <b>Description</b>                                                                                                                                            |
|----------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>NFR-1</b>   | <b>Usability</b>                  | The application should provide a simple, responsive, and user-friendly interface for booking metro tickets and recharging smart cards.                        |
| <b>NFR-2</b>   | <b>Security</b>                   | Passenger information, payment details, and booking records must be securely stored using ServiceNow security features with role-based access control (RBAC). |
| <b>NFR-3</b>   | <b>Reliability</b>                | The system should accurately process ticket bookings, recharge requests, and QR code generation without data loss or duplication.                             |
| <b>NFR-4</b>   | <b>Performance</b>                | Ticket booking, fare calculation, Flow Designer execution, and QR code generation should complete within a few seconds for a seamless user experience.        |
| <b>NFR-5</b>   | <b>Availability</b>               | The application should be available 24×7 to allow passengers to book tickets and recharge smart cards anytime.                                                |
| <b>NFR-6</b>   | <b>Scalability</b>                | The system should support a large number of concurrent users and booking transactions during peak travel hours without affecting performance.                 |